1.Input: Unstructured military news documents

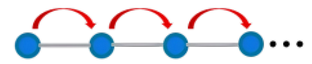
Document	Text Content
Doc 1	During the operation, the Type A ICBM was launched from Base B toward the Eastern Maritime Test Corridor.
Doc 2	Earlier this week, the Ministry of National Defense of Country M announced a test operation involving the new Type A ICBM.
Doc 3	According to intelligence sources, the missile eventually impacted within Test Zone C, demonstrating extended-range capability.

Legend (Semantic Annotation)

- Trigger
- Entity
- Location
- Time

2. Limitations of Existing Methods

Existing Methods: Isolated Modules Without Unified Spatiotemporal Modeling or Constraints

Module	Modeling Granularity	Main Method	Main Method	Illustration
Event Extraction	Only identifies single events and arguments	Sequence labeling	Cannot link cross-document events	
Temporal Relation Extraction	Cross-document alignment is difficult	Event-pair classification / ranking	Implicit temporal cues; unstable event ordering	
Location Argument Extraction	Isolated location tags	Sequence labeling / classification	Lacks path semantics	

Result: Incomplete Event Chain

- ✗ Difficult event alignment; fragmented chains
- ✗ Missing spatial-path semantics
- ✗ Inaccurate long-span boundary detection

3. Core Challenges

Distinctive characteristics of military news Pose Challenges to Existing Methods

Challenge Dimension	Description	Example / Feature
Difficult event alignment and fragmented event chains	Events for the same target are scattered across multiple documents; temporal order is implicit, making alignment difficult	Events are reported at different times and by different sources; their order must be inferred
Continuous Evolution of Spatial Paths	Target movement follows an "Origin → Via → Destination" path rather than isolated locations	A military target moves from A to B to C, forming a continuous path
Long Arguments and Cross-Sentence Expressions	Entity descriptions are long and highly modified; boundaries are ambiguous and cross-sentence dependencies are strong	Complex modifiers, cross-sentence references, and nested expressions
Large Task Heterogeneity and Inconsistent Modeling Granularity	Boundary detection, path modeling, and role classification require different granularities, making shared-parameter learning difficult	High-frequency simple roles differ substantially from low-frequency complex arguments

4. Research Objectives

Research Goal: Construct Spatiotemporally Consistent Structured Event Chains

Requirements
✓ Consistent cross-document event alignment
✓ Clear and coherent temporal ordering
✓ Complete and traceable spatial paths
✓ Accurate long-argument boundaries and complete semantics

Event ID	Trigger	Time	Argument	Previous Event	Event Chain	Spatial Path
E1	launched	During the operation	Type A ICBM	E2	E2 → E1 → E3	Base B → Eastern Maritime Test Corridor
E2	test operation	Earlier this week	Ministry of National Defense of Country M, Type A ICBM	--		Country M
E3	impacted	eventually	Type A ICBM	E1		Base B → Eastern Maritime Test Corridor → Test Zone C